



The powder is not the price. So what does a medicine's price **pay for?**

Last post: the active ingredient got cheaper for decades. This post opens the rest of the ledger — **eight pharmaceutical companies' own SEC filings**, human and animal health side by side. The answer explains a forty-year price divergence.

Two industries, one code.

Opposite anatomies.

R&D · SHARE OF
REVENUE

**19% vs
7%**

Human pharma
median **19.4%**;
animal health **7.3%**
— a **2.7×** gap. The
human price buys
science.

COST OF GOODS ·
SHARE OF REVENUE

**30% vs
45%**

Inverted: human
29.6%, animal
44.6%. The animal
price buys
manufacturing —
which is where the
ingredient lives.

WHAT REMAINS AFTER
COGS+R&D+SG&A

**~30 vs
~23**

Per \$100 of revenue:
human ~30, animal
~23. **Similar
discipline,
opposite routes to
it.**

And this is why Part 1 looked the way it did: a P&L built on R&D must reprice to fund the next molecule — human prices rose +75% real over twenty years. A P&L built on manufacturing tracks its inputs — veterinary prices fell 4% real on the same window. **The cost anatomy predicted the price behaviour.**

Straight from the filings. Nothing modelled.



Source

Annual 10-K figures via **SEC EDGAR's XBRL API** — machine-readable, public, keyless. CIKs resolved from SEC's own ticker file at fetch time.



Measure

COGS, R&D and SG&A as % of revenue — **mean of FY2023-25** per company, **median** across each group so no single filer drags the figure.



Two clean groups

Human: Pfizer, Lilly, AbbVie, Bristol Myers Squibb. **Animal:** Zoetis, Elanco, Phibro — pure-play filers only. Merck (~9% animal health) is shown but **pooled with neither group**.

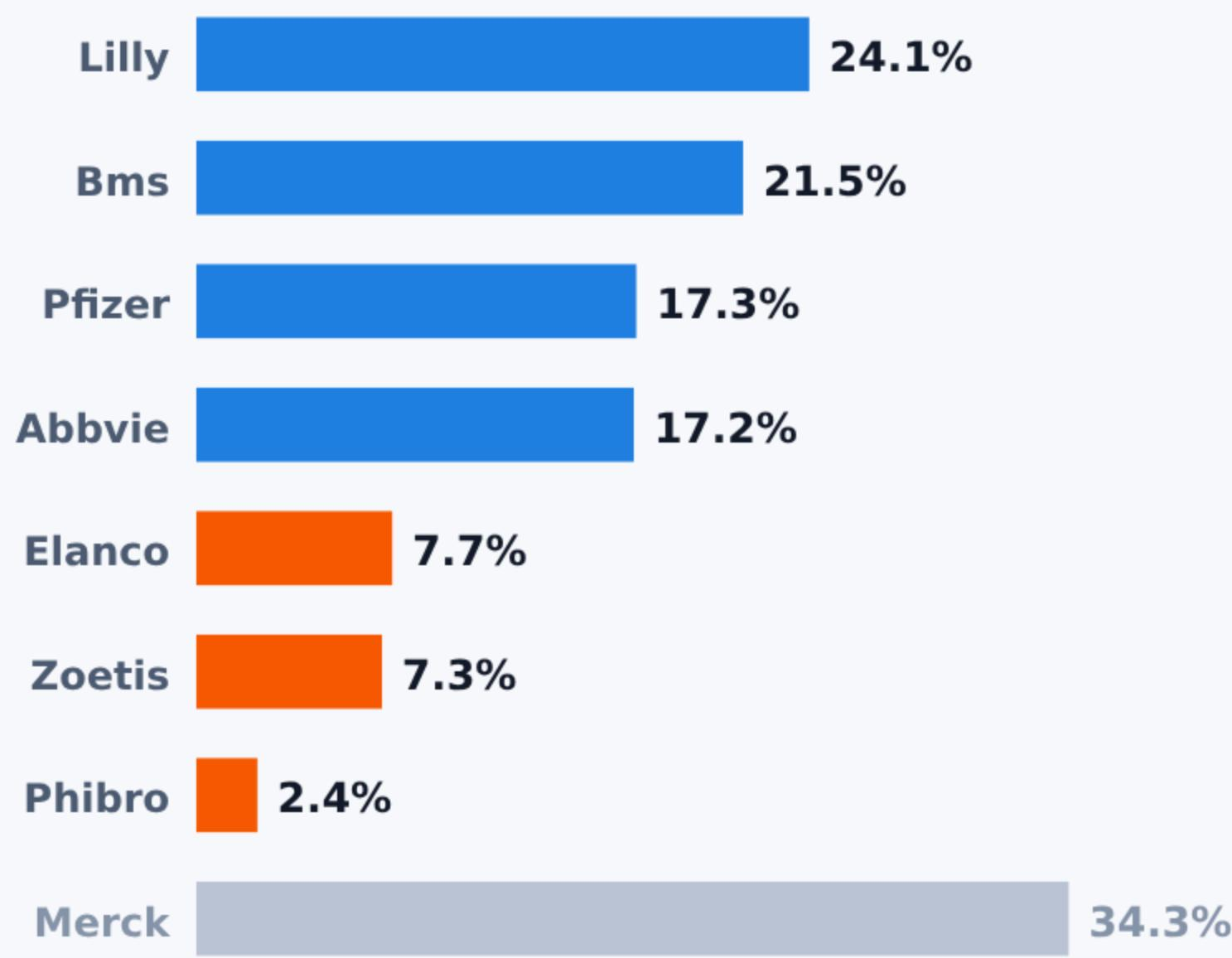


What COGS is not

COGS is not the ingredient. No filer splits API out of its cost of goods — the ingredient from Part 1 sits *inside* these bars, unquantified.

The human price buys science.

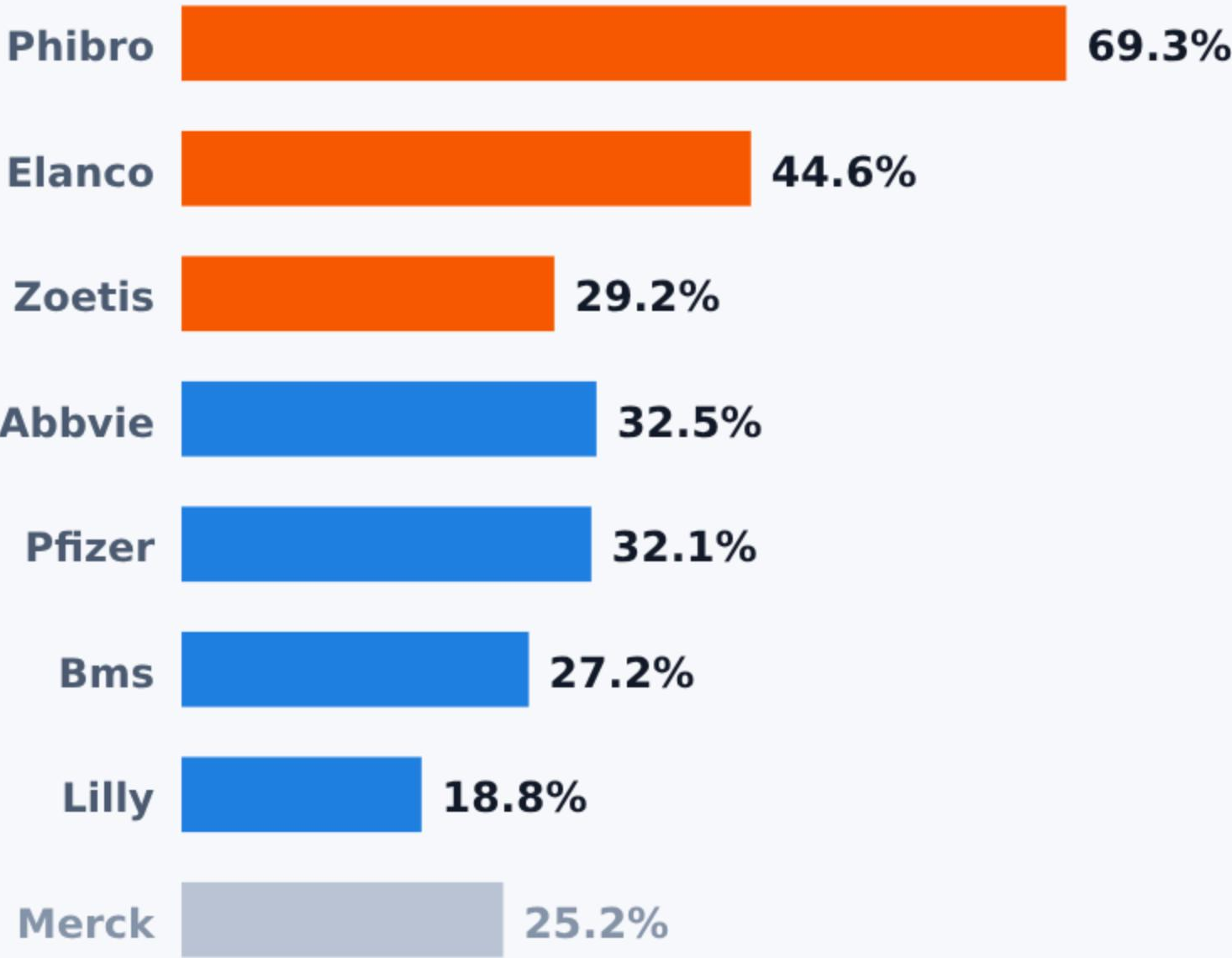
R&D as a share of revenue, FY2023–25 average. Every human filer sits between 17% and 24%. Every animal filer sits **below 8%**.



SEC 10-K FILINGS, FY2023–25 AVG · MERCK'S 34.3% INCLUDES LARGE ACQUIRED-IPR&D CHARGES – ONE REASON IT POOLS WITH NEITHER GROUP

The animal price buys manufacturing.

Cost of goods as a share of revenue — the ranking flips. Animal health runs **29-69%**; human pharma 19-33%. **The heavier the COGS bar, the more the Part 1 ingredient story is your story.**

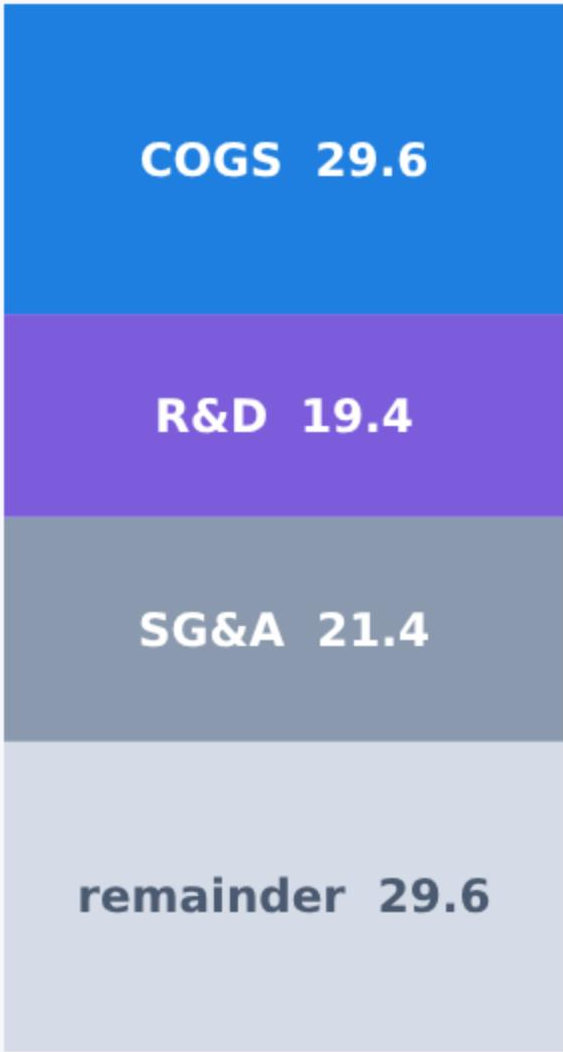


SEC 10-K FILINGS, FY2023-25 AVG · SPREAD INSIDE ANIMAL HEALTH IS REAL: PETCARE-HEAVY ZOETIS 29% VS NUTRITION-HEAVY PHIBRO 69%

Anatomy of \$100 of revenue.

Group medians, side by side. Same industry code, **opposite construction** — and a similar residual at the end of the line.

HUMAN PHARMA · MEDIAN OF 4



ANIMAL HEALTH · MEDIAN OF 3



MEDIANS: HUMAN N=4, ANIMAL N=3 · "REMAINDER" = 100 - COGS - R&D - SG&A: OPERATING CUSHION BEFORE OTHER LINES, NOT PROFIT

Two anatomies. Two price histories.

R&D-BUILT P&L · HUMAN

+75%

Real price change of US human-dominated pharma over the clean 20-year window (2001–2021).

Science must be repriced — every new molecule is funded by the last one's price.

MANUFACTURING-BUILT P&L · VETERINARY

–4%

Real price change of the US **veterinary** preparations index on the same window (BLS WPU0634).

Manufacturing tracks its inputs — and its inputs got cheaper.

The takeaway for anyone reading animal-health prices through a human-pharma lens: the benchmark is wrong. Veterinary medicine has priced like the manufacturing business it is — for forty years. Both facts come from public data and both survive the deflator.

Read before **quoting.**

Small samples, stated

Four human filers, three animal. Medians, not means; three fiscal years averaged. This is an anatomy sketch, not a census.

Global, not US-only

10-K figures are worldwide consolidated. Nothing here is a US-market cost share.

Heterogeneity is real

Inside animal health, COGS runs 29% (Zoetis) to 69% (Phibro) — petcare and nutrition are different businesses sharing a label.

Accounting seams

Filers switched revenue/R&D tags mid-history (unioned, logged); acquired-IPR&D inflates some R&D lines (Merck). No headcount tag exists in XBRL — employee costs stay unmeasured.

Same ingredient market. Different businesses built on it.

IF YOU BENCHMARK

Benchmark within the anatomy — an animal-health R&D or COGS ratio judged against human-pharma norms is a category error.

IF YOU BUY

The heavier the COGS bar, the more input discipline pays — procurement and FX cadence are margin levers, not admin.

IF YOU SELL

Vet pricing history is a fairness story — four decades without out-pricing inflation is worth saying out loud.

Every source

Filings · SEC EDGAR XBRL company facts (data.sec.gov), 10-K FY values, FY2023–25: Pfizer, Eli Lilly, AbbVie, Bristol Myers Squibb · Zoetis, Elanco, Phibro · Merck (shown, not pooled). Concept tags per company recorded in the repository.

Price histories · BLS PPIs via FRED — veterinary preparations WPU0634, pharmaceutical preparation industry PCU325412325412, CPI-U deflated, continuous-segment analysis.

Part 1: the input ledger — 44 years of the active ingredient. Every figure in both posts is recomputed from data committed in the repository, and each carries its named source.